Laspeyre's price index number $P_{01}^{L} = \frac{\sum p_1 q_0}{\sum p_0 q_0} X 100$

Paasche's price index number $P_{01}^{P} = \frac{\sum p_1 q_1}{\sum p_0 q_1} X 100$

Fisher's price index number $P_{01}^{F} = \sqrt{P_{01}^{L} \; x \; P_{01}^{P}} = \sqrt{\frac{\sum p_1 q_0 \;\; X \sum p_1 q_1}{\sum p_0 q_0 \;\; X \sum p_0 q_1}} \; X 100$

- Time Reversal Test : $P_{01} \times P_{10} = 1$. Factor Reversal Test : $P_{01} \; X \; Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$
- Cost of Living Index Number

  Aggregate Expenditure Method = $\frac{\sum p_1 q_0}{\sum p_0 q_0} X 100$. Family Budget Method = $\frac{\sum PV}{\sum V}$
- Causes of Variation

  1. Chance Causes (or) Random causes 2. Assignable Causes
- Control Charts

  (i) Centre Line (CL) indicates the desired standard level of the process.

  (ii) Upper Control Limit (UCL) indicates the upper limit of tolerance.

  (iii) Lower Control Limit (LCL) indicates the lower limit of tolerance.

The control limits for $\overline{X}$ chart in two different cases are

case (i) when $\overline{\overline{X}}$ and SD are given	case (i) when $\overline{\overline{X}}$ and SD are not given
$UCL = \overline{\overline{X}} + 3\frac{\sigma}{\sqrt{n}}$	$UCL = \overline{\overline{X}} + A_2 \overline{R}$
$CL = \overline{\overline{X}}$	$CL = \overline{\overline{X}}$
$LCL = \overline{\overline{X}} - 3\frac{\sigma}{\sqrt{n}}$	$LCL = \overline{\overline{X}} - A_2 \overline{R}$

The control limits for R chart in two different cases are

case (i) when SD are given	case (i) when SD are not given
$UCL = \overline{R} + 3\sigma_R$	$UCL = D_4 \overline{R}$
$CL = \overline{R}$	$CL = \overline{R}$
$LCL = \overline{R} - 3\sigma_R$	$LCL = D_3 \overline{R}$

XII Std - Business Maths & Stat EM Chapter 9.indd 235 2/28/2020 5:11:42 PM